Follow-the-Expert: Expert-Preserving Demonstration Synthesis in a Gaussian-Splat Twin

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Abstract: A 3D Gaussian Splatting (3DGS) twin can turn one real demonstration into hundreds of rendered ones and train a policy that runs on the real robot. We find that for skills defined by their path, this Real2Sim2Real loop works best when the synthesized motion stays the expert's. Follow-the-Expert (FTE) replays each phase of one VR demonstration as a dynamic movement primitive, moves only the phase boundaries and their object splats, and lets the splat's density steer the gripper around obstacles. On a real Boston Dynamics Spot, FTE policies succeed in 75.6% of rollouts on four shape-sensitive tasks, against 40.0% for trajectory optimization and 3.8% for keyframe motion planning; all three tie at 91.3% where only the end state matters. FTE also leads with obstacles absent from the demonstration (86.7% vs. 60.0% on three tasks), and doubling the offset bounds needs only more rendered demonstrations.

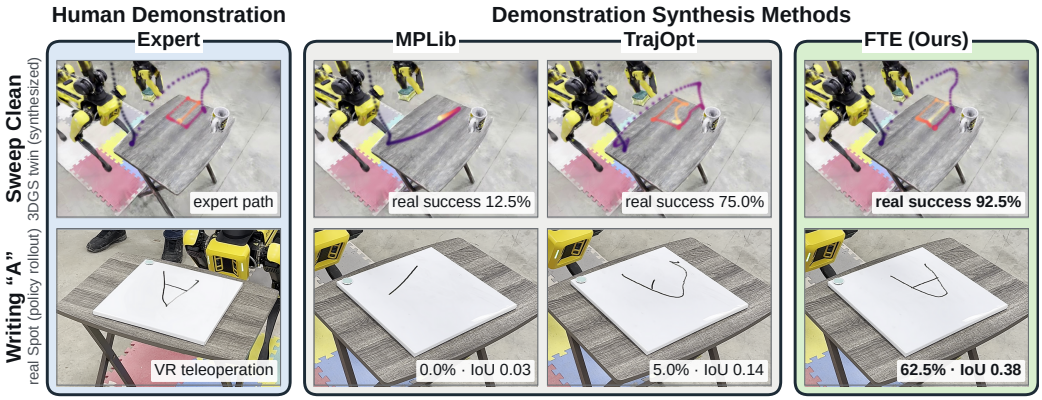


Figure 1: **Keeping the expert's motion is what transfers to the real robot.** Top: Sweep Clean demos synthesized in the 3DGS twin (end-effector path, dark to bright; Expert: the demo replayed in the twin) and each generator's real-robot success. Bottom: Writing "A" on the real Spot after one rollout per policy (Expert: the VR demonstration); success and mean stroke IoU over 40 rollouts.

1 Keep the expert, move the boundaries

A splat twin makes demonstrations cheap; keeping the expert makes them right. 3DGS reconstructions [1] train real-robot policies on rendered data [2, 3, 4]. One-demonstration generators in this family retarget keyframes and plan or optimize the motion between them (RoboSplat [5], 1001 DEMOS [6]; we re-implement them as our baselines MPLib, keyframes plus motion planning, and TrajOpt, keyframes plus trajectory optimization), which suits pick-and-place but replaces or distorts the expert's path when the path is the task (Fig. 1). Segment-based generators keep object-centric segments found from contact [7, 8, 9]; FTE keeps every phase, including contact-free ones such as a camera scan, and uses the splat as a repulsive potential rather than a collision map [10].

Late-Breaking Results (non-archival), 10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

	No obstacles			With obstacles absent from the demonstration					
	Success ↑, 256 demos			Success ↑, 256		Success ↑, 500		Collision ↓, 500	
Task	MPLib	TrajOpt	FTE	TrajOpt	FTE	TrajOpt	FTE	TrajOpt	FTE
Sweep Clean	12.5	75.0	**92.5**	67.5	**80.0**	82.5	**95.0**	22.5	**7.5**
Push–Pick–Place	2.5	30.0	**75.0**	30.0	**50.0**	35.0	**82.5**	60.0	**12.5**
Writing "A"	0.0	5.0	**62.5**	–	–	–	–	–	–
Videography	0.0	50.0	**72.5**	42.5	**60.0**	62.5	**82.5**	27.5	**22.5**
Sponge Pick–Place	**95.0**	92.5	92.5	82.5	**85.0**	92.5	**95.0**	**2.5**	**2.5**
Stove	87.5	**90.0**	**90.0**	77.5	**80.0**	95.0	**97.5**	5.0	**2.5**

Table 1: **Real-robot success and collision rates (%), 40 rollouts per cell**; one policy per cell, trained on 1 real + 256 or 500 synthetic demos. Shape-sensitive tasks above the rule; bold: best per group; –: not run. Success by inspection (Writing: two judges, 85–100% agreement); collision: visible contact. MPLib plans without obstacle geometry, as in RoboSplat (10.0–17.5% at 500 with a 3DGS proxy).

FTE moves the boundaries and replays the motion. One handheld scan of the lab (reconstructed in 5–10 min, shared by all tasks) is registered to the robot by ICP (7.7 mm trimmed-mean residual); Spot's own splat, split into per-link Gaussians, is posed by forward kinematics. One VR demonstration is split at operator button presses into phases, each fitted with Cartesian and orientation dynamic movement primitives (DMPs) [11]. During synthesis we sample bounded offsets of the phase boundaries ($1\times$ bounds, about 0.1 m per axis), move the linked object splats with them, and roll out each DMP with the expert's forcing term [12], keeping the approach, contact path, gripper timing, and orientation. Near obstacles, the summed density ρ of designated obstacle Gaussians (an uncalibrated proxy) pushes the end effector along $-\nabla\rho$ before the rollout rejoins the expert's path [13]. Rendered front and top views plus the real demo train a Diffusion Policy [14]; the twin simulates no dynamics.

2 Three findings on a real Spot

1. Keeping the path decides the shape-sensitive tasks. Four tasks are shape-sensitive: wiping a spill (Sweep Clean), pushing a block before picking it (Push–Pick–Place), writing "A", and filming a flower along an arc (Videography, six phases); Sponge Pick–Place and Stove are endpoint controls. Only the generator differs: all methods share the twin, demonstration, policy, and 40 test configurations inside the $1\times$ bounds. FTE leads TrajOpt on every shape-sensitive task (Table 1), 121/160 rollouts against 64/160 (MPLib 6/160; task-stratified Cochran–Mantel–Haenszel (CMH) test, $p < 10^{-10}$), and all three reach 91.3% on the endpoint controls. Path-preserving warps match FTE on Sweep Clean (affine 87.5%, spline 92.5%): the path is what matters, and DMPs give obstacle avoidance an acceleration-level entry point.

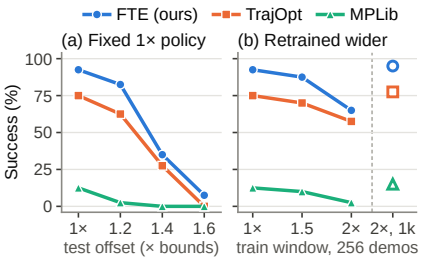


Figure 2: **Range costs rendering.** Sweep Clean, N=40 per point. (a) $1\times$ policies tested farther out. (b) Retrained on wider windows; hollow: $2\times$ with 1,000 demos.

2. The splat that renders the views also clears obstacles. With an obstacle absent from the demo, FTE leads TrajOpt on the three shape-sensitive tasks at both budgets (86.7% vs. 60.0% at 500 demos, CMH $p < 10^{-5}$) and collides less on each; without the density term, FTE drops to 0–17.5% on Sweep, Push–Pick–Place, Sponge and Stove.

3. Range costs rendering, not teleoperation. Beyond its bounds a fixed $1\times$ policy degrades with the baselines (Fig. 2a), since FTE retargets locally. Retrained on a $2\times$ window, 256 demos spread too thin (Sweep 65.0%, Sponge 67.5%); 1,000 reach 95.0% and 97.5%, back at the $1\times$ level (Fig. 2b).

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